

July 13, 2026 – Custom Mobile Dashboard & Live Bluetooth Telemetry

Over the past four days, I completed the development of a custom Flutter mobile application for the RC Crawler Telemetry System and successfully integrated it with the ESP32 using Bluetooth Low Energy (BLE). I began by designing a landscape-oriented dashboard optimized for monitoring the crawler while driving. The interface includes analog gauges for vehicle speed and motor RPM, along with dedicated displays for battery voltage, motor temperature, pitch, and roll.

After designing the user interface, I implemented Bluetooth Low Energy communication within the Flutter application. This required creating a dedicated Bluetooth service to automatically discover and connect to the ESP32, receive telemetry notifications, and continuously update the dashboard with live sensor data. Following several rounds of debugging and testing, I successfully established a reliable wireless connection between the mobile application and the telemetry system.

The mobile dashboard now receives and displays live telemetry data, including:

- Battery Voltage
- Motor Temperature
- Wheel RPM
- Vehicle Speed
- Pitch
- Roll

I also improved the overall user interface by redesigning the dashboard to resemble a modern automotive instrument cluster. This included animated analog gauges, Bluetooth connection status indicators, battery and temperature displays beneath each gauge, and a Recovery Mode button reserved for future functionality. The application was successfully tested on my iPhone, where it displayed real-time telemetry updates while communicating wirelessly with the ESP32.

With this milestone complete, the telemetry system now consists of a fully functional embedded sensor platform, wireless Bluetooth communication, and a custom mobile application capable of displaying live vehicle telemetry in real time. The next phase of the project will focus on designing a custom KiCad PCB to replace the breadboard prototype, allowing the ESP32, sensors, and power circuitry to be permanently integrated into the RC crawler.